EMPLOYEE ENGAGEMENT, SATISFACTION, AND BURNOUT DIAGNOSTIC ANALYSIS

Research Paper | Employee HR Analytics Dataset

Important Dataset Note

The supplied CSV contains 1,470 records and 31 columns. Its structure resembles a commonly circulated IBM HR Analytics employee attrition dataset, but the file does not prove that the records belong to Palo Alto Networks. Treat the results as sample-data analysis unless the original source is verified.

Abstract

This project presents a descriptive framework for examining employee engagement, satisfaction, work-life balance, overtime, career stage, and attrition context. Four ordinal ratings—job involvement, job satisfaction, environment satisfaction, and relationship satisfaction—are combined into an engagement score from 0 to 100. A rule-based workload indicator combines overtime and low work-life balance. These are exploratory indicators, not validated clinical burnout measures or individual attrition predictions.

1. Introduction

Employee experience can influence collaboration, productivity, innovation, and retention. Monitoring multiple indicators can help HR teams identify areas for further investigation. This project examines satisfaction and work conditions, especially overtime, work-life balance, business travel, tenure, job role, and attrition status.

2. Problem Statement and Objectives

The objective is to build a unified descriptive view of engagement and workload-related signals and identify groups that may warrant further investigation. A low rating does not prove burnout, and these indicators should not be used alone for employment decisions.

Objectives: construct an engagement index; summarize work-life balance; calculate a transparent overtime/balance diagnostic; compare departments, roles, overtime status and tenure; examine attrition context; and recommend supportive HR actions.

3. Dataset Description

The CSV contains 1,470 rows and 31 columns. Fields include demographics, department, job role, satisfaction ratings, overtime, travel, income, tenure, career progression, and attrition. Before final submission, validate the dataset source, missing values, duplicates, rating ranges, and meaning of the attrition field.

4. Data Validation and Preparation

Trim whitespace from column names and text values. Validate satisfaction and work-life balance ratings against the expected 1–4 range. Convert numeric fields safely, normalize overtime labels, and map attrition values consistently. Calculate engagement only when at least three of four component ratings are available. Report missingness and never interpret missing values as positive outcomes.

5. Methodology and KPI Definitions

Engagement Index: mean of JobInvolvement, JobSatisfaction, EnvironmentSatisfaction, and RelationshipSatisfaction when at least three ratings are present.

Engagement Score: Engagement Index divided by 4, multiplied by 100. Illustrative categories are Low below 60, Moderate from 60 to below 80, and High at 80 or above. These thresholds are project-defined, not validated organizational standards.

Work-Life Balance Index: WorkLifeBalance divided by 4, multiplied by 100.

Burnout-related diagnostic score: 50 points for overtime and 50 points for a work-life balance rating of 1 or 2. Scores of 100, 50, and 0 are labelled High, Medium, and Low. This is a rule-based workload proxy, not a clinical assessment. Missing balance data can understate risk.

Satisfaction stability and workload stress are exploratory indicators. Consistent satisfaction ratings can be consistently low or high, and an overtime/travel index is not a validated stress measure.

6. Preliminary Descriptive Results

Preliminary calculations on the supplied CSV produced an average engagement score of approximately 68.1 out of 100 among records with a calculable score. Approximately 28.3 percent of records were marked as overtime, and 126 records met the project's high-risk rule. These are descriptive results for the supplied dataset and are not verified findings about Palo Alto Networks. Re-run the analysis after validating the scoring rules and data.

7. Interpretation of EDA

Review engagement score distributions, satisfaction dimension averages, overtime comparisons, work-life balance patterns, and differences by department, job role, job level, and tenure. Show group sizes with averages and report denominators for percentages. Group differences are descriptive and do not prove causation.

8. Recommendations

1. Review staffing, deadlines, on-call rotation, and recovery time where overtime and low work-life balance overlap.

2. Prioritize repeated patterns in adequately sized groups instead of publishing named employee risk lists.

3. Validate dashboard patterns through pulse surveys, manager check-ins, focus groups, and voluntary feedback.

4. Review career paths, internal mobility, mentoring, and promotion criteria where long promotion gaps coincide with lower engagement.

5. Explore whether travel or commute demands coincide with work-life balance issues in specific groups.

6. Evaluate aggregate measures before and after interventions while accounting for changes in team composition.

7. Restrict access to employee-level data, suppress small groups, and avoid using exploratory scores as the sole basis for employment decisions.

9. Limitations

Dataset provenance is unverified and the data must not be represented as internal Palo Alto Networks data without evidence. The dataset may be historical, synthetic, anonymized, or non-representative. Satisfaction ratings are ordinal; averaging them assumes approximate equal spacing. Engagement bands and burnout-related scores are project-defined rather than validated instruments. Missing data and group composition may bias comparisons. Associations do not establish causation or reliably predict individual departures. The dataset does not directly measure exhaustion, psychological safety, actual workload hours, or intent to stay.

10. Conclusion

This project demonstrates how employee HR data can be transformed into an engagement index, workload-related diagnostic indicators, and group comparisons. The dashboard can help frame questions around overtime, work-life balance, career development, and employee experience. It should support further investigation and humane interventions, not serve as a definitive burnout diagnosis or individual attrition prediction. Verify data provenance before making company-specific claims.

11. Tools and References

Pandas documentation: https://pandas.pydata.org/docs/

NumPy documentation: https://numpy.org/doc/

Plotly Python documentation: https://plotly.com/python/

Streamlit documentation: https://docs.streamlit.io/

Verify the original dataset source and license before final submission.